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Key Points:

- We present a novel method for RFI detection and mitigation based on cyclic spectrum analysis and compressive statistical sensing
- Experiments with synthetic radar-based signals have been conducted, and accurate detection is achieved
- Arecibo Observatory L-band data contaminated with impulsive narrowband RFI from surrounding active radars are also used to demonstrate technique feasibility

Supporting Information:

- Supporting Information S1

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Radio Frequency Interference Detection and Mitigation Using Compressive Statistical Sensing

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Abstract Quasiperiodic radio frequency interference (RFI), such as those generated by telecommunication and active radar systems, is commonly encountered in radio astronomy observations. Such RFI-contaminated signals contain hidden periodicities due to cyclic features involved in their formation (e.g., carrier frequencies, periodic keying of the amplitude, and baud rates). RFI signal characterization and its subsequent excision based on the well-known cyclic spectrum analysis have been previously demonstrated; however, the high complexity of the algorithm and the computational cost of its implementation have limited its utility in radio astronomy, rendering less sophisticated solutions. To overcome this challenge, we present a novel method for RFI detection and mitigation based on efficient estimation of the cyclic spectrum by compressive statistical sensing (CSS) of sub-Nyquist data. CSS performs second-order statistical estimation such as cyclic spectrum using a reduced number of input samples, thereby enabling accelerated performance. To validate the feasibility of the proposed method, we conduct experiments with simulated data and assess the detection and mitigation results under different parameter settings, for example, interference-to-noise ratio, additional RFI sources, frequency resolution, and input data size. We demonstrate the real performance of the method by analyzing radio astronomy data (~1.3 GHz) acquired with the L-wide band receiver at the Arecibo Observatory, which is typically corrupted by active air surveillance radars located nearby. Our CSS-based solution enables robust and efficient detection of the RFI frequency bands present in the L-band data, and subsequent excision by blanking is also demonstrated.

1. Introduction

The past few decades have witnessed the proliferation of man-made transmission systems and their concomitant spectral leakage into protected bands for scientific purpose. Use of the radio spectrum is coveted by both active services (e.g., cellular, radar, and wireless internet) and passive services (e.g., radio astronomy and Earth exploration satellite service). A well-known example is the spurious infringement, by long-range air surveillance radar systems and GPS transmissions, into the L-band at 1400–1427 MHz, which is allocated exclusively for passive radio astronomy and Earth exploration satellite service sensing of soil moisture, pulsar signatures, and arguably the most crucial spectral feature in radio astronomy: the 21-cm (1420 MHz) atomic hydrogen emission line (Akyildiz et al., 2011; Bhat et al., 2005; Camps et al., 2000). Meanwhile, scientific interest in the spectrum outside of designated research bands has intensified with the advent of wide bandwidth receivers and spectrometers, which enable higher sensitivity for continuum observations and extended frequency coverage for spectroscopic studies. However, such hardware advances have rarely been realized to their full potential due to long-standing challenges in isolating natural emissions from near-ubiquitous radio frequency interference (RFI) from anthropogenic sources.

Since most radio signals of scientific interest are extremely faint, the effects of RFI contamination can be disproportionate, ranging from amplifier saturation and thus total data corruption to increased uncertainty due to lowered sensitivity. For example, impulsive RFI may mask or mimic signatures of pulsars or other short-duration transient radio bursts of cosmic origins whose detection is an important science goal for current state-of-the-art telescopes such as the Allen Telescope Array, the Low Frequency Array and the Square Kilometer Array, or the single-dish Green Bank Telescope and Arecibo Observatory (AO), among others. On the other hand, continuous, broadband RFI from sources such as windmills, electric fences, power

lines, or digital radio and video broadcasting, can further reduce the achievable sensitivity, bias antenna calibration, and ruin usable data from affected stations (Fridman & Baan, 2001).

Mitigating RFI contamination of scientific data is predicated first and foremost on its robust identification. Contemporary schemes for RFI detection and mitigation have been successful to some degree, particularly for strong signals whose localization in time, space, or frequency is well characterized. However, current limitations in spectral sensing and signal classification capabilities usually lead to insufficient interference cancellation, inability to adapt to rapid changes in the interference signal, partial removal of wanted data, or excessive computational requirements for postprocessing of the accumulated data. Meanwhile, weak RFI, with its subtle yet insidious impact on data quality, is notoriously difficult to recognize, much less suppress, using conventional spectrum sensing approaches. A classification of these current RFI detection and mitigation techniques is presented in Offringa (2012) and includes the following: using spatial information provided by multifeed systems together with coherence subtraction methods to nullify effects of the RFI (Ellingson & Hampson, 2003; Leshem & van der Veen, 2000a; Kocz et al., 2010), removing the RFI through physical installation of fixed reference antennae and subsequent correlation processing (Barnbaum & Bradley, 1998), and blanking out RFI-related high values in the time-frequency domain based on thresholding techniques (Baan et al., 2004; Leshem & van der Veen, 2000b; Weber & Faye, 1998).

Single-dish radio telescopes are particularly sensitive to RFI because both incoming signals of interest and RFI enter the receiver coherently. In this context, several methods have been developed for this specific scheme, which include RFI modeling and matching, RFI source prediction, and adaptive filters. For example, Ellingson and Hampson (2003) describe a method to detect impulse-like RFI generated by air surveillance active radars in AO L-band measurements, which is based on match filtering, that is, the convolution of an a priori estimated pulse model and the acquired data, with a subsequent thresholding step to determine RFI presence. Furthermore, radar echoes from aircraft are considered additional sources of RFI that directly affect observations of neutral hydrogen and hydroxyl molecule spectral lines at the single-dish Green Bank Telescope. Dong et al. (2005) developed a method to predict the spatial and temporal location of such RFI sources based on Kalman filter tracking of aircraft positions. Pulsar detection at the single-dish Parkers Radio Telescope, Australia, has been affected by the installation of digital TV transmitters near the telescope facilities. Kesteven et al. (2005) made use of Adaptive Noise Cancelling techniques (Widrow & Stearns, 1985) to nullify effects of RFI in the signal of interest. Adaptive Noise Cancelling methods require the presence of an auxiliary reference antenna (Fridman & Baan, 2001) or a reference multifeed system (Barnbaum & Bradley, 1998; Briggs et al., 2000) that detects the radio interference while remaining insensitive to the astronomical signal.

AO (whose measurements are examined in this manuscript) hosts numerous receivers for wide-band astronomical observations but has no auxiliary antennas and is strongly contaminated by periodic RFI generated from nearby air surveillance radars, particularly in the L-wide band (1.15–1.74 GHz). In an effort to deal with such strong RFI, the National Astronomy and Ionosphere Center developed a device for real-time mitigation of a strong ground-based aviation radar located at Pico del Este, Puerto Rico. This device, also known as “RFI blanker,” tracks a predefined pattern of the timing between pulses for this particular radar and then blanks the receiver output in a time window around the expected pulse arrival times $\sim 400 \mu\text{s}$. Since the window duration is ~ 33.3 times longer than the average pulse duration, numerous samples of the signal of interest are lost (Douglas, 2011; Ellingson & Hampson, 2003). To address this problem, hardware-independent techniques that analyze statistical properties of RFI signals—such as spectral kurtosis or cyclic spectrum—and accurately determine their temporal occurrence have received considerable attention in radio astronomy.

Spectral kurtosis, which measures the peakedness of a probability distribution, is useful in problems concerned with the detection of transients in noisy signals as well as identification of non-Gaussian signals (Antoni, 2004; Dwyer, 1984; Vrabie et al., 2003). In radio astronomy, spectral kurtosis has become a common tool for real-time detection and removal of non-Gaussian RFI sources (Nita et al., 2007). On the other hand, cyclic spectral analysis (CSA) deals with cyclostationarity processes that feature inherent temporal periodicities, as exhibited in telecommunications and active radar signals, associated with carrier frequencies, symbol rates, or coding patterns (Gardner, 1988, 1991; Gardner et al., 2006). CSA has been frequently used in the telecommunications field, specifically in cognitive radio networks, where cyclostationarity-based feature detection aims to identify the signal modulation types and separate interferences from licensed primary

signals (Lunden et al., 2009). Some applications of CSA in radio astronomy include the real-time CSA detectors for identification of periodic RFI contamination introduced by satellite transmissions into data acquired by the single-dish Nancay Radio Telescope, France (Bretteil & Weber, 2004; Weber & Faye, 1998), and the CSA-based RFI detection technique based on array signal processing for the Low Frequency Array Radio Telescope, Netherlands (Hellbourg et al., 2012). The continuing need for accelerated detection algorithms in telecommunications has led the development of cutting-edge signal processing techniques focused on the estimation/reconstruction of the cyclic spectrum using a compressive sensing framework, where a reduced number of input samples are used to estimate this second-order statistical property (see section 2; Tian, 2011; Tian et al., 2012).

In this paper, we extend CSA techniques toward the real-time detection of RFI generated by active radars, which contaminate measurements at single-dish radio telescopes like the Arecibo Radio Observatory. The approach developed in this work relies on compressive statistical sensing (CSS), a technique designed for the recovery of statistical properties of random processes (Romero et al., 2016). Specifically, this work presents a cyclostationary-based RFI blanker that includes a CSS-based estimator for the cyclic spectrum and a methodology for RFI detection and blanking. Section 2 describes the assumed signal model, the CSS algorithm, and the methodology for RFI detection and subsequent blanking in the time-frequency domain. In section 3, we assess the performance and robustness of our proposed algorithm using simulated data. Additionally, we show the algorithm results using a representative example of real data acquired with the Arecibo Radio Observatory L-wide receiver and contaminated with strong active radar RFI from Punta Boriquem Air Surveillance Radar.

2. Compressive Cyclic Feature Detection

In this section we describe the cyclic feature detection algorithm, which is convenient for statistical RFI detection in radio astronomy (Cucho-Padin et al., 2018; Waldrop et al., 2017). In order to reduce the signal acquisition cost under wide-band scenarios (i.e., high sampling frequencies to acquire and analyze an extensive frequency region), we implement a sparse cyclic spectrum recovery approach, which leverages the strong sparsity over the two-dimensional (2-D) cyclic spectrum. Given the recovered sparse 2-D cyclic spectrum, the frequency occupancy of the RFI signals can be identified via a band-by-band multicycle generalized likelihood ratio test (GLRT).

2.1. Signal Model

Let $x(t)$ be a continuous-time wide-band received signal in the frequency range $[-f_{\max}, f_{\max}]$. There are P RFI sources emitting over this wideband, where the p th RFI signal is denoted by $x_p(t), p \in \{1, 2, \dots, P\}$. Then, the signal corrupted by additive noise $w(t)$ is given by

$$x(t) = \sum_{p=1}^P x_p(t) + w(t). \quad (1)$$

Suppose that RFI signals $x_p(t)$ are nonoverlapping in frequency. Then, the task of RFI detection is to identify the frequencies on which these RFI signals are present.

2.2. Cyclostationarity

Cyclostationarity of a nonstationary random process is a second-order statistical feature that reveals the inherent periodicity of some stochastic signals (Gardner, 1991). Since man-made RFI signals contain periodical patterns in their formation such as carrier frequencies, periodic keying of the amplitude, or baud rates, exploiting such cyclostationarity can help detect these RFI signals which are buried in strong but stationary noise.

To review the cyclostationarity, the cyclic spectrum $S(\alpha, f)$ of $x(t)$ is defined as

$$S_x(\alpha, f) = \int R_x(\alpha, \tau) e^{-j2\pi f \tau} d\tau, \quad (2)$$

where f is the frequency, α is the cyclic frequency, τ is the time delay, and $R_x(\alpha, \tau)$ is the cyclic autocorrelation function defined as

$$R_x(\alpha, \tau) = \int R_x(t, \tau) e^{-j2\pi\alpha t} dt, \quad (3)$$

where $R_x(t, \tau) = E\{x(t+\tau/2)x^*(t-\tau/2)\}$ is the autocorrelation function equivalent to the autocovariance for a zero-mean signal. The cyclic spectrum $S_x(\alpha, f)$ is also defined as the spectral correlation function, which essentially indicates the correlation of spectral components that are separated by α in the frequency domain. It exhibits nonzero values only at a discrete set of cyclic frequencies as $\alpha = k/T_0, k \in \mathbb{Z}$, which reveals the inherent second-order periodicity of $x(t)$. That is, $R_x(t, \tau)$ is periodic with period T_0 as $R_x(t, \tau) = R_x(t+T_0, \tau)$. In this sense, the periodicity arisen from the underlying mechanism of signals such as carrier frequency, symbol rate, and modulation type can be captured via CSA.

2.3. Sparse Cyclic Spectrum Recovery

Cyclic feature estimation typically demands sampling at a higher-than-Nyquist rate in order to induce cyclostationarity and generally requires a long sensing time to acquire reliable cyclic statistics (Dandawate & Giannakis, 1994; Gardner, 1991). Such requirements present significant difficulties due to extremely high acquisition costs in the wide-band regime and limited-length observation windows in dynamic environments. To overcome such challenges, we make use of a compressive cyclic feature detection approach that is efficient in processing wide-band signals and effective in collecting statistical characteristics.

Our approach is motivated by a unique sparsity property of the 2-D cyclic spectrum. Besides the sparsity in frequency domain due to the limited frequency occupancy of narrowband RFI signals embedded within wide-band radio astronomy observations, an additional dimension of sparsity arises owing to the cyclostationarity-dependent structure in cyclic frequency domain. Such a 2-D cyclic spectrum allows stronger sparsity in the cyclic-frequency-and-frequency (α, f) biplane, which therefore motivates the recovery of the sparse cyclic spectrum from compressive samples.

The linear compressive sampling of continuous $x(t)$ in equation (1) can be described in the matrix-vector form as

$$\mathbf{z}_t = \mathbf{A}\mathbf{x}_t, \quad (4)$$

where $\mathbf{x}_t$ is an $N \times 1$ vector representing the discrete version of $x(t)$ with the frequency resolution of $2f_{\max}/N$, the $M \times N$ matrix $\mathbf{A}$ denotes a compressive sampler such as a random fat Gaussian matrix with a compression ratio $M/N < 1$, which forms an undetermined system, and $\mathbf{z}_t$ is the obtained compressive samples. The covariance $\mathbf{R}_z = E\{\mathbf{z}_t \mathbf{z}_t^H\}$ in the compressed domain and the covariance $\mathbf{R}_x = E\{\mathbf{x}_t \mathbf{x}_t^H\}$ in the uncompressed domain have the linear relationship as

$$\text{vec}\{\mathbf{R}_z\} = \text{vec}\{\mathbf{A}\mathbf{R}_x\mathbf{A}^H\} = (\mathbf{A}^* \otimes \mathbf{A}) \text{vec}\{\mathbf{R}_x\}, \quad (5)$$

where $\otimes$ denotes Kronecker product, the operator $\text{vec}\{\cdot\}$ concatenates all columns of a given matrix to form a vector, and super indices H and $*$ refer to the Hermitian transpose (or conjugated transpose) and the conjugate operation, respectively.

From equations (2) and (3), $S_x(\alpha, f)$ is a double Fourier transform of $R_x(t, \tau)$, which then leads to the linear relationship between their digital counterparts $\mathbf{S}_x$ and $\mathbf{R}_x$ as

$$\text{vec}\{\mathbf{R}_x\} = \text{vec}\{\mathbf{F}^{-1} \mathbf{S}_x \mathbf{F}^{-H}\} = ((\mathbf{F}^{-H})^T \otimes \mathbf{F}^{-1}) \text{vec}\{\mathbf{S}_x\}, \quad (6)$$

where $\mathbf{F}$ is the N -point discrete Fourier transform matrix and super index T indicates a transpose operation. Substituting equation (6) into equation (5), we can express the vector form covariance of the compressive samples as a linear function of the vector form cyclic spectrum as

$$\mathbf{r}_z = \mathbf{T} \mathbf{s}_x, \quad (7)$$

where $\mathbf{r}_z = \text{vec}\{\mathbf{R}_z\}$, $\mathbf{T} = (\mathbf{A}^* \otimes \mathbf{A})((\mathbf{F}^{-H})^T \otimes \mathbf{F}^{-1})$, and $\mathbf{s}_x = \text{vec}\{\mathbf{S}_x\}$.

Given the vector form covariance $\mathbf{r}_z$ that is usually available in the form of its sample covariance calculated over L snapshots (where a snapshot is an N length of the input data), $\hat{\mathbf{r}}_z = \text{vec}\{(1/L) \sum_{l=1}^L \mathbf{z}_t(l) \mathbf{z}_t^H(l)\}$. In

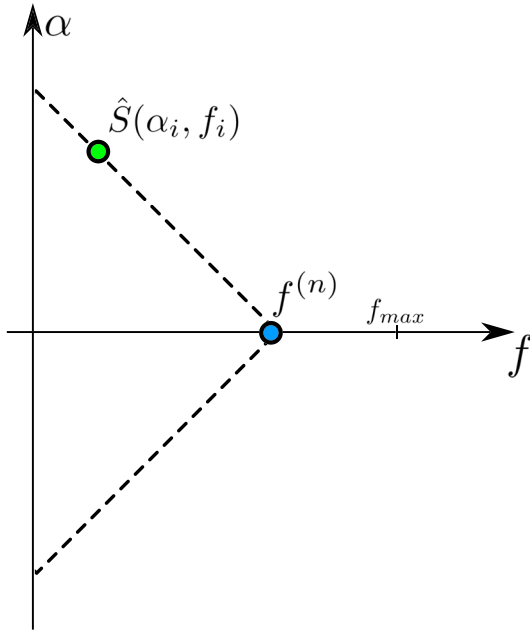


Figure 1. Band-by-band detection using the estimated 2-D cyclic spectrum $\hat{S}_x$. For a given $f^{(n)}$ (restricted to the f axis and presented with a blue dot), a semidiamond (dashed line) is defined by equation (9). Discrete values described by $\{\hat{S}_x(\alpha_i, f_i)\}_{i \in \mathcal{J}_d^{(n)}}$ (green dot) are stacked and generate the vector $\hat{\mathbf{c}}^{(n)}$, which is used to determine the presence or absence of radio frequency interference.

practice, the cyclic spectrum recovery problem amounts to solving $\mathbf{s}_x$ in equation (7), which is an underdetermined linear system with M^2 equations and N^2 unknowns with $M < N$. To solve this problem, we make use of the strong sparsity of the 2-D cyclic spectrum, which can be incorporated into the recovery of cyclic spectrum by penalizing the ℓ_1 -norm of the $\mathbf{s}_x$ to induce a sparse solution. Thus, we formulate an unconstrained ℓ_1 -norm regularized least squares problem for reconstructing $\mathbf{s}_x$ as

$$\hat{\mathbf{s}}_x = \underset{\mathbf{s}_x}{\operatorname{argmin}} \|\mathbf{r}_z - \mathbf{T}\mathbf{s}_x\|_2^2 + \lambda \|\mathbf{s}_x\|_1, \quad (8)$$

where $\|\mathbf{x}\|_1$ stands for the sum of the absolute values of the components of $\mathbf{x}$ and λ is a weighting scalar that governs the trade-off between the sparsity of the solution induced by the ℓ_1 -norm term and the residual error reflected by the ℓ_2 -norm least squares term.

2.4. Band-by-Band Detection

After recovering the sparse 2-D cyclic spectrum in its vectorized form $\hat{\mathbf{s}}_x$ through equation (8), the goal of RFI detection is to identify the specific frequency locations $f^{(n)}$ on which RFI signals are present, where $f^{(n)} = 2f_{\max}n/N, \forall n \in [-N/2, N/2]$ and $n \in \mathbb{Z}$. This task amounts to a band-by-band inspection of the spectrum occupancy by RFI signals over the entire frequency range.

On a single band $f^{(n)}$, if an RFI signal is present, it would occupy a semidiamond region $\mathcal{J}^{(n)}$ of the 2-D cyclic spectrum map defined by the cyclic frequency and frequency pairs (α, f) satisfying

$$\begin{aligned} \text{if } f^{(n)} > 0 & : f + \frac{|\alpha|}{2} = f^{(n)} \text{ and } 0 \leq f + \frac{|\alpha|}{2} \leq f_{\max}, \\ \text{if } f^{(n)} < 0 & : f - \frac{|\alpha|}{2} = f^{(n)} \text{ and } -f_{\max} \leq f - \frac{|\alpha|}{2} < 0, \end{aligned} \quad (9)$$

The geometric structure presented in equation (9) considers complex input signals that would produce an asymmetric spectrum in the f frequency dimension. Thus, we stack the estimated $\{\hat{S}_x(\alpha_i, f_i)\}_{i \in \mathcal{J}_d^{(n)}}$ into a vector $\hat{\mathbf{c}}^{(n)}$ of length $|\mathcal{J}_d^{(n)}|$, where $\mathcal{J}_d^{(n)}$ is the discrete set containing the indices of pairs (α_i, f_i) satisfying equation (9) (see Figure 1). In this sense, $\hat{\mathbf{c}}^{(n)}$ is formed by selected entries of $\hat{\mathbf{s}}_x$ in equation (8) as $\hat{\mathbf{c}}^{(n)} = \mathbf{J}_n \hat{\mathbf{s}}_x$, where the selection matrix $\mathbf{J}_n \in \{0, 1\}^{|\mathcal{J}_d^{(n)}| \times N^2}$ is obtained from the $N^2 \times N^2$ identity matrix by keeping its rows with indexes $\forall i \in \mathcal{J}_d^{(n)}$ only.

To test the presence of an RFI signal at the frequency $f^{(n)}$, the following binary hypothesis test is formulated

$$H_1 : \mathbf{c}^{(n)} = \mathbf{c}^{(n)} + \boldsymbol{\epsilon} \quad H_0 : \mathbf{c}^{(n)} = \boldsymbol{\epsilon} \quad (10)$$

where $\mathbf{c}^{(n)}$ represents the true vector of cyclic spectrum values and $\boldsymbol{\epsilon}$ is asymptotically Gaussian distributed; that is, $\lim_{LN \rightarrow \infty} \sqrt{LN} \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \Sigma^{(n)})$, where $\Sigma^{(n)}$ is the asymptotic covariance matrix. Because $\Sigma^{(n)}$ is not readily available, a blind estimator for $\Sigma^{(n)}$ is used given the available compressive samples $\{\mathbf{z}_l(l)\}_{l=1}^L$ (Tian et al., 2012). Replacing $\Sigma^{(n)}$ with $\hat{\Sigma}^{(n)}$ results in a data-adaptive GLRT detector. Treating $\mathbf{c}^{(n)}$ as an unknown nuisance parameter, the adaptive GLRT can be derived for equation (10), which yields the following test statistic:

$$\mathcal{T}^{(n)} = (\mathbf{c}^{(n)})^H (\hat{\Sigma}^{(n)})^{-1} \mathbf{c}^{(n)}. \quad (11)$$

Specifically, $\mathcal{T}^{(n)}$ is compared with a threshold to make decisions on the RFI presence at frequency $f^{(n)}$ and this threshold is chosen to comply with the allowable false alarm rate (Trees, 2001). This is a multicycle

GLRT, because $\hat{\mathbf{c}}^{(n)}$ contains multiple cyclic frequencies α_i for $i \in \mathcal{F}_d^{(n)}$. Finally, putting together all the bands occupied by RFI signals, we can detect the occupancy of RFI signals over the entire wide band.

2.5. Methods to Solve ℓ_1 -Regularization Problems

Estimation of the 2-D cyclic spectrum $\hat{\mathbf{s}}_x$ relies on the search for the optimal solution of the convex problem defined in equation (8), where the ℓ_1 -norm regularization term enforces a desired sparsity within the estimate values. Several convex solvers that deal with this type of regularization can be found in the literature, including interior point methods (Ben-Tal & Nemirovski, 2001), subgradient strategies *Shooting* (Fu, 1998) and *Grafting* (Perkins et al., 2003), and gradient projection algorithms (Beck & Teboulle, 2009; Daubechies et al., 2004; Figueiredo et al., 2007). Gradient Projection or Proximal Gradient methods have demonstrated a better time efficiency due to the fewer matrix-vector multiplications involved in their formulation. One popular example is the Iterative Shrinkage-Thresholding Algorithm (ISTA) where each iteration involves matrix-vector multiplication followed by a shrinkage/soft-threshold step. In addition, ISTA has a complexity of $\mathcal{O}(1/k)$, where the term $1/k$ is the error converge rate function and the variable k is the number of iterations (see details in Daubechies et al., 2004). Beck and Teboulle (2009) have developed the Fast-ISTA (FISTA), a method that accelerates the original ISTA and achieves a complexity of $\mathcal{O}(1/k^2)$; thus, error convergence is achieved k times faster.

With the advent of high-performance parallel architectures such as graphical processing units and Field-Programmable Gate Array, accelerated implementations can be attained to the extent that algorithms mainly rely on matrix-vector operations as well as on shorter iteration cycles in those recursive sections. In addition, reduction of the processing time plays an essential role in RFI detection and mitigation when data collection campaigns span several days.

2.6. RFI Detection and Mitigation Procedure

RFI detection and mitigation procedure is composed of several steps depicted in Figure 2. An analog signal coming from the radio telescope is downconverted to baseband and then discretized with a sampling frequency $F_s > 2B$, where B is the attainable signal bandwidth $[-B/2, B/2]$ around the zero frequency. The resulting samples are the in-phase and quadrature complex components ($x[m] = I[m] + jQ[m]$, $x \in \mathbb{C}$, $j = \sqrt{-1}$ and m is the discrete-time variable) and are the initial input of our proposed technique. First, a K -length window ($K = N \times L$, see section 2.3) from the complex signal $x[m]$ is normalized and used as the input for the CSS-based method, which will estimate the correspondent 2-D cyclic spectrum $\hat{\mathbf{S}}_x$. Second, the band-by-band detection will perform the summation of the elements within a semidiamond pattern for a given $f^{(n)}$ and generate the vector $\hat{\mathbf{c}}^{(n)}$. Third, the test statistic is calculated as explained in section 2.4 to generate the vector $\mathcal{T}^{(n)}$. Fourth, we will consider that an RFI is present in a frequency bin (n) if $\mathcal{T}^{(n)} > \xi$, where the threshold value ξ is selected to achieve a desired probability of false alarm (p_{fa}). Finally, the resulting binary vector is used to perform the mitigation process. Strategies for RFI mitigation are divided into two main areas: blanking, where the contaminated data are deleted or labeled (to be avoided in a further analysis), and canceling, where RFI is removed while leaving astronomical data unaffected. In this study, our focus is to demonstrate the capability of the CSS-based algorithm to determine the RFI occupancy in the 2-D cyclic spectrum of contaminated data; therefore, we implement a “blanker” that will identify and label the cyclostationarity RFI in the time-frequency (t, f) plane. Future work could focus on the implementation of a canceling strategy based on specific RFI models as described by Ellingson and Hampson (2003).

3. Numerical Experiments

In this section, we demonstrate the feasibility of the RFI detection and mitigation algorithm described in section 2 using two different approaches. First, as described in section 3.1, we establish bounds of detection fidelity using synthetic data that mimics the quasiperiodic RFI signatures of active radars at varying levels of interference to signal ratio (ISR). We conduct additional tests of the algorithm by superposing this synthetic RFI with BPSK interference. For each set of synthetic signals, we apply the CSS-based algorithm to detect and blank contaminated frequencies in the time-frequency domain. Receiver operating characteristic (ROC) curves describing the probability of detection (p_d) in terms of the probability of false alarm (p_{fa}) are calculated and depicted as a means of defining accuracy of the binary hypothesis test (equation (10)). Next, as described in section 3.2, we apply our algorithm to actual L-band data acquired by the single-dish

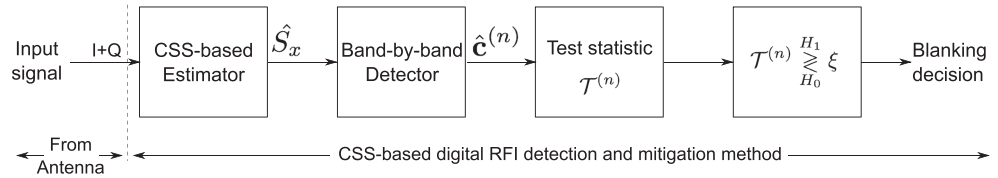


Figure 2. radio frequency interference (RFI) detection and mitigation algorithm based on compressed statistical sensing (CSS). The input of our algorithm are the complex components I and Q of the signal $x[m]$ provided by the antenna receiver. The CSS-estimator calculates the 2-D cyclic spectrum $\hat{S}_x$. Then, the band-by-band detector generates the vector $\hat{\mathbf{c}}^{(n)}$ by adding elements in a diamond pattern for each $\mathbf{f}^{(n)}$. Next, test statistic $\mathcal{T}^{(n)}$ is calculated from $\hat{\mathbf{c}}^{(n)}$ as explained in section 2.4. Finally, a threshold ξ is selected in order to achieve a desired probability of false alarm (p_{fa}) and is used together with $\mathcal{T}^{(n)}$ to determine the presence of a RFI signature.

AO L-wide receiver, which contains quasiperiodic RFI generated by nearby air surveillance radars in addition to the desired astronomical signal. We demonstrate the blanking mitigation process on this real data in the (t, f) plane.

3.1. RFI Detection and Mitigation With Simulated Data

We first demonstrate the feasibility of our RFI detection and mitigation technique using a synthetic time domain signal constructed as the superposition of cyclostationary RFI and an astronomical signal of interest, which we model as having a white Gaussian distribution in frequency. The variable ISR governs the ratio of RFI to astronomical signal power. AO L-band data are well known to be strongly contaminated by active radar signals, which usually have pulse-like and frequency-sweep patterns (Cucho-Padin et al., 2018; Ellingson & Hampson, 2003). Such pulses have an approximate duration of 100 μ s, a typical ISR > 10 dB, and a frequency-sweep rate $\Delta f \approx |0.5/50|$ (MHz/ μ s). We generate a synthetic frequency-sweep RFI signal, or chirp, as follows:

$$x[n] = e^{j2\pi F_1 \cos(2\pi F_2 \frac{n}{F_s})}; \quad (12)$$

where F_s is the sampling frequency, F_1 is the rate of change in frequency (the chirp rate), and the expression $\cos(2\pi F_2 \frac{n}{F_s})$ defines a time-varying and zero frequency-crossing structure with a period $1/F_2$. The synthetic signal (RFI + Gaussian data), which is used as input to the RFI detection and mitigation algorithm described above, is sampled at $F_s = 5$ MHz and filtered to have the range $[-0.5, 0.5]$ MHz. Figure 3a depicts the time-frequency (t, f) domain representation of several chirps spanning 600- μ s duration of the input signal for ISR = 10 dB.

We apply the CSS algorithm (see section 2) to the time domain simulated data in order to identify the cyclic feature presented in the spectral content of the RFI and thus accurately determine the time-varying range of uncontaminated frequencies. For time-varying RFI detection, we use a K -length time window shifting over the input data sequence. Within each time window, the K -length data segment is reformed as an $N \times L$ data matrix with $K = NL$; that is, a K -length data segment is composed of L subsegments or snapshots, each of which has N data samples. Here, the choice of value K reflects a trade-off between the finite sample effect for averaging and the capability of tracking time-varying RFI signals. Given fixed N , a large K leads to a large L that yields more snapshots for the calculation of sample covariance at the cost of requiring more time for each detection. On the other hand, a small K enables fast detection by means of quick response with an inevitable loss of accuracy in sample averaging.

The input data are first divided into overlapping K -length segments that are initially normalized, and each segment includes L snapshots. At each snapshot, the compressive samples are collected with the compression ratio M/N , resulting in a $M \times L$ input matrix. Next, the CSS algorithm uses that matrix to recover the cyclic spectrum $\hat{S}_x(\alpha, f)$, which is performed by using the FISTA algorithm. Then, the band-by-band detection algorithm outputs a detected vector $\hat{\mathbf{c}}^{(n)}$ as an estimate of the principal frequency components within each time window. Subsequently, test statistic $\mathcal{T}^{(n)}$ is calculated as explained in section 2.4. A ground truth spectral content vector $\mathbf{g}^{(n)}$ associated with each window is generated with a priori information such as Δf . Lastly, to demonstrate the detection performance, we depict the receiver operating characteristic (ROC) curves

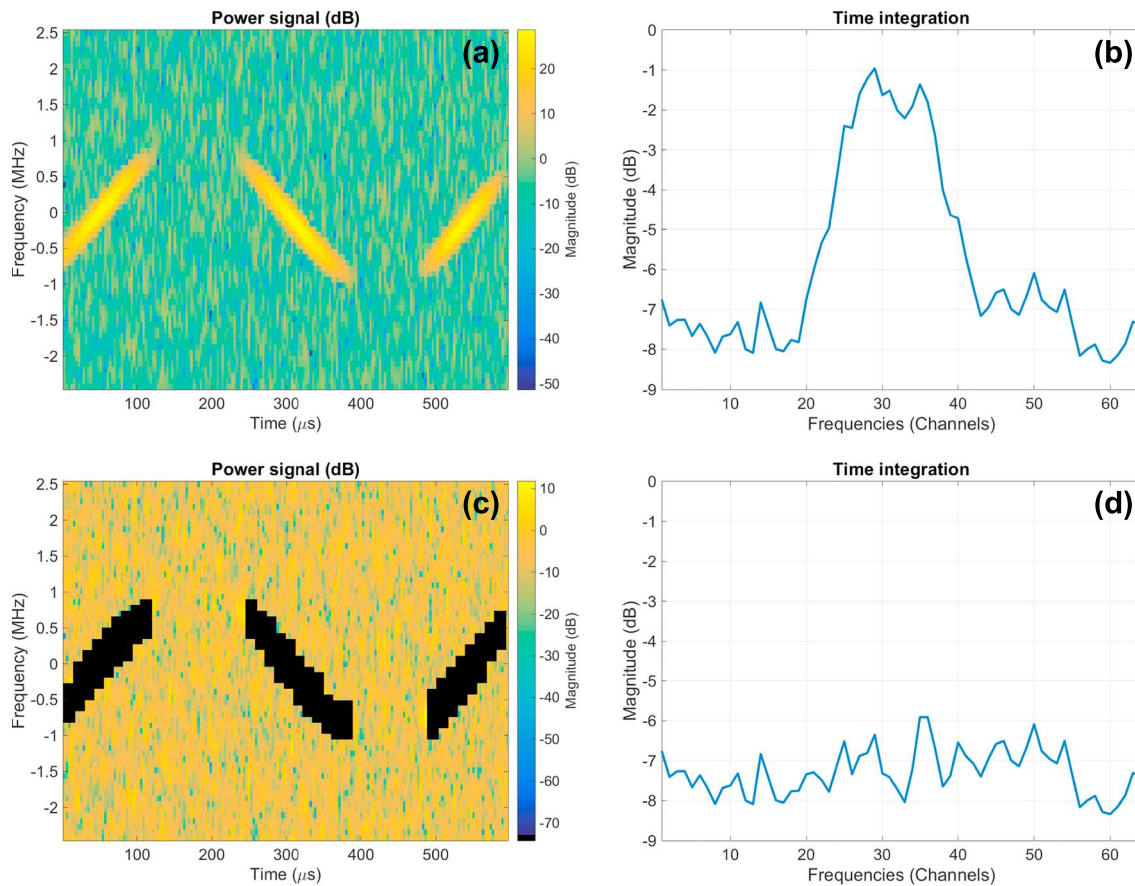


Figure 3. (a) Time-frequency domain spectrogram of a 600- μ s segment of the synthetic input signal (chirp radio frequency interference [RFI] + Gaussian “data”). (b) Time integration of the input signal shown in panel (a). (c) Spectrogram of input signal shown in panel (a) after RFI detection and blanking, where black regions indicate the presence of RFI. (d) Time integration of the blanked signal shown in panel (c).

based on both $\mathcal{T}^{(n)}$ and $\mathbf{g}^{(n)}$. We evaluate our algorithm using three compression rate values $M/N=\{1,0.75,0.5\}$ (where 1 corresponds to uncompressed data), two window sizes defined by $K=192,160$, with $N=64,32$ and $L=3,5$ respectively, and three different values for $\text{ISR} = 3, 5$, and 10 dB. In addition, the overlap is set to 33.3% for $K=192$ and 40% for $K=160$ to avoid a high computational load. With the given configurations, ROC curves presented in Figures 4 and 6 are generated with ~ 200 data segments for $K=192$, $N=64$, and with ~ 260 data segments for $K=160$, $N=32$. Several compression rates (M/N) have been thoroughly evaluated in the context of cognitive radio applications by Tian et al. (2012) where modulated “telecommunication” signals are detected based on CSS methodology. This experimental study reveals that for higher compression rates $M/N \leq 0.25$ the probability of detection p_d drops drastically from ~ 0.8 to ~ 0.4 (see Figure 4 of the referred manuscript). Therefore, we adopted compression rates greater than $M/N > 0.25$ and specifically $M/N = 0.5$ as maximum boundary in our analysis.

Figure 4 depicts ROC curves for the specified parameter settings. As expected, superior performance is achieved with larger ISR and larger compression ratio, and uncompressed data allow for excellent RFI detection ($p_d > 0.9$ for $p_{fa} < 0.1$) for all ISR and data window sizes investigated. However, in all of the 18 individual cases, the probability of detection (p_d) is greater than 0.8 for a probability of false alarm (p_{fa}) less than 0.2, which indicates the good detection fidelity of the algorithm under both low intensity ISR and high compression conditions. Reduction in window size, and thus the number of signal samples used as input, from $K=192$ (Figures 4a, 4c, and 4e) to $K=160$ (Figures 4b, 4d, and 4f) is found to yield a lower probability of detection for a given probability of false alarm for large ISR in particular; on the other hand, the algorithm converges on a given level of RFI detection faster when fewer samples are used as input.

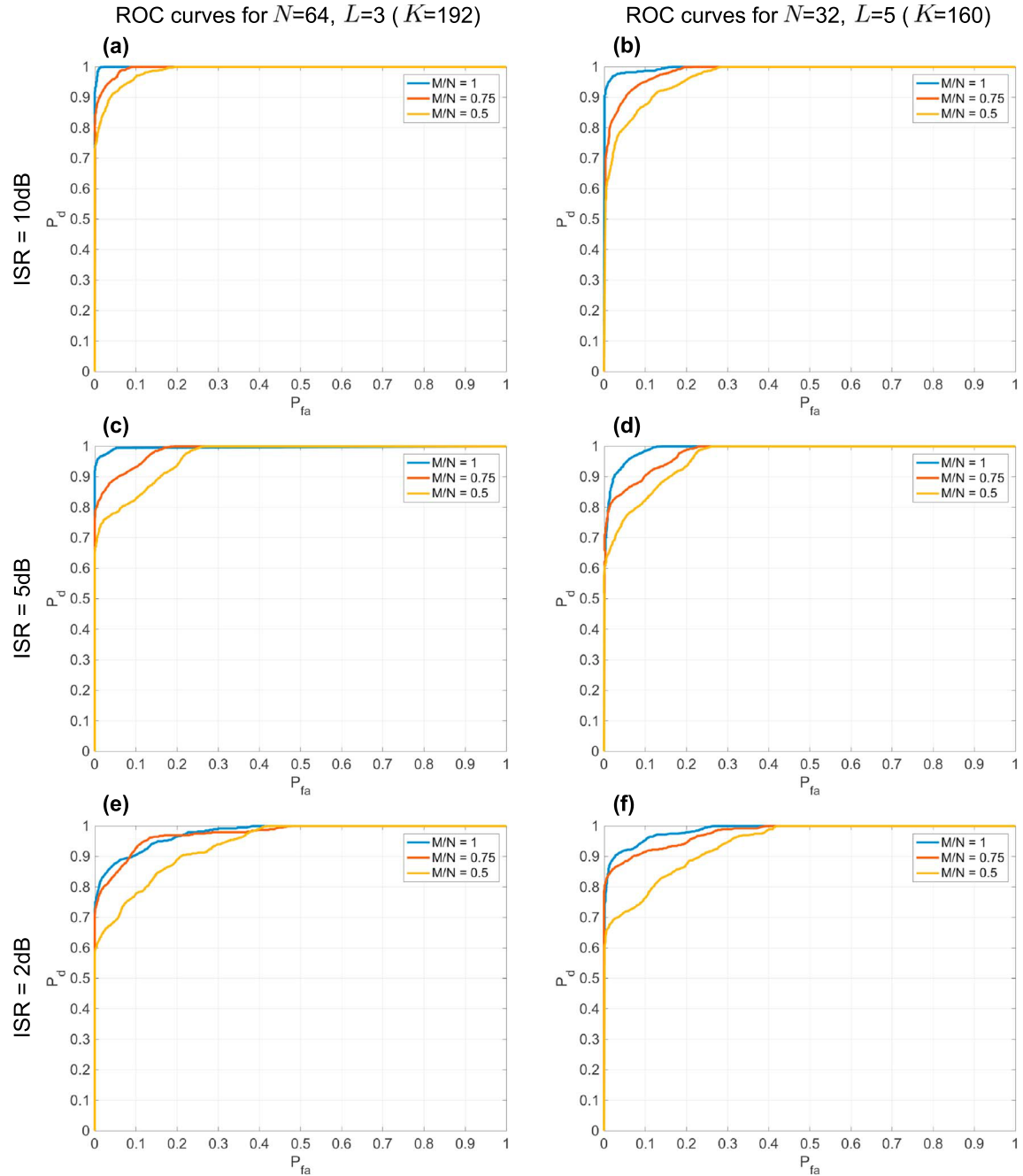


Figure 4. ROC curves for the compressive statistical sensing-based cyclic feature detector with different parameter settings and three different compression rates M/N . The input signal is the complex chirp with additive white Gaussian noise that is depicted in Figure 3a. (first column) ROC curves for $N=64$ and $L=3$ with (a) $ISR = 10$ dB (c) $ISR = 5$ dB, and (e) $ISR = 2$ dB. (second column) ROC curves for $N=32$ and $L=5$ with (b) $ISR = 10$ dB, (d) $ISR = 5$ dB, and (f) $ISR = 2$ dB. ISR = interference to signal ratio; ROC = receiver operating characteristic.

Further tests with additional ISR , N , and L configurations are required for definitive assessment regarding the bounds on and conditions for technique validity. However, these 18 examples suggest that reliable detection can be performed with the proposed algorithm for a wide range of realistic sampling conditions, yielding a high probability of detection ($p_d \geq 0.9$) for a probability of false alarm $p_{fa}=0.2$ when $ISR \geq 5$ dB.

For blanking of the detected RFI signal, we set $p_{fa}=0.2$, which yields a threshold $\xi=0.05$. Each element (frequency bin) of test statistic vector $\mathcal{T}^{(n)}$ is then compared with ξ to generate a binary vector that indicates the

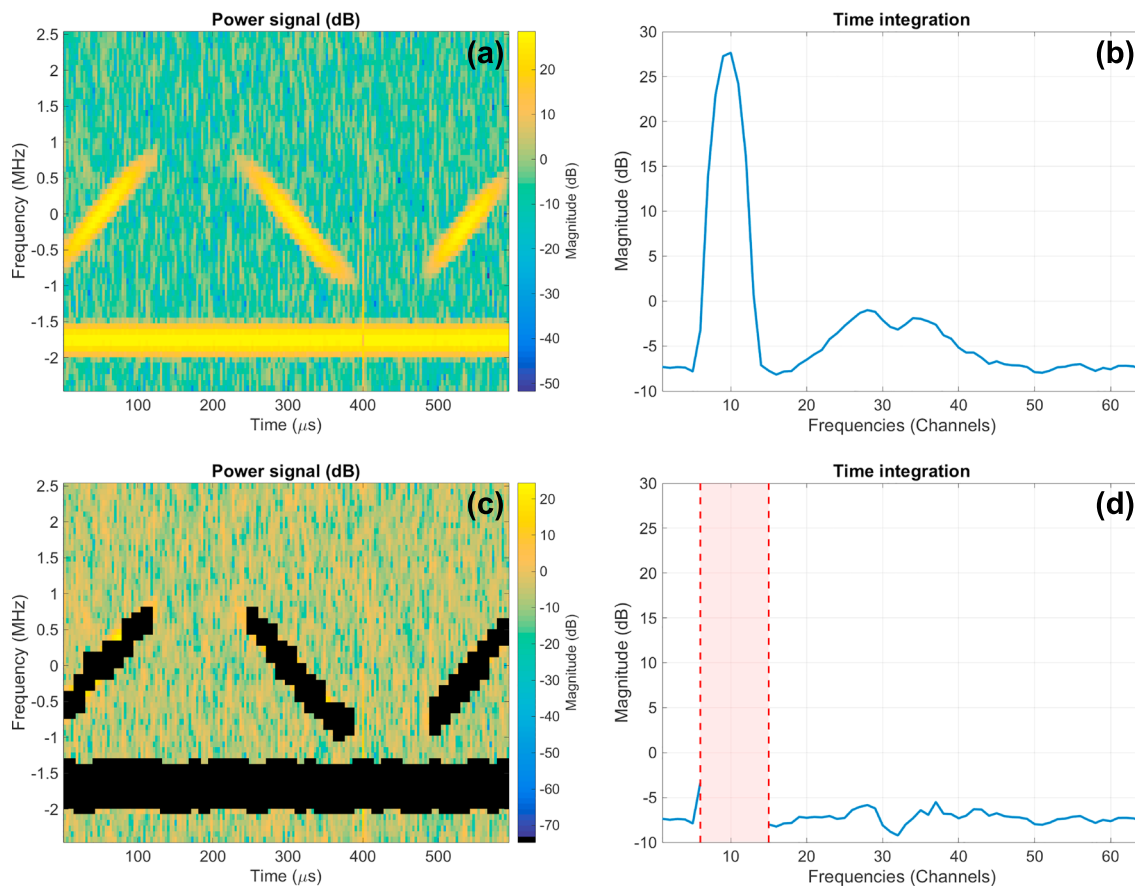


Figure 5. Spectrograms and time integrations of a 600- μ s segment of the synthetic input signal (chirp RFI + BPSK RFI + Gaussian “data”), before and after RFI detection and blanking. Format follows Figure 3. The algorithm determined the signal to be fully contaminated over frequencies associated with the time-continuous BPSK interference (shaded region in panel d). RFI = radio frequency interference.

presence or absence of RFI. Figures 3a and 3b depict a 600- μ s-length portion of the simulated data (panel a) and its temporal integration (panel b). For this example, we have selected $N=64$ and $L=3$ (i.e., $K=192$) and sliding window overlap of 33%, or 64 samples. The spectrogram in Figure 3a was generated by calculating the 64-point short-time Fourier transform with an overlap of 50%; therefore, the $K=192$ data samples used by our algorithm determine the presence or absence of RFI for every five columns of the spectrogram. Figure 3b depicts the time integration of the spectrogram and is calculated by averaging all the power values in a given row (i.e., frequency). This graph is useful for radio astronomers, since they can evaluate the impact of the RFI over the signal of interest in the frequency domain. The frequency channel range [1,64] is associated with frequency values $[-2.5 \text{ MHz}, 2.5 \text{ MHz}]$ and presents a power concentration in the central frequency channels due to the presence of the chirp RFI signal. The spectrogram in Figure 3c shows the results of the blanking process, while Figure 3d depicts its temporal integration.

We conduct a second investigation of our algorithm performance using the original synthetic signal composed of the RFI chirp and Gaussian “astronomical data,” now additionally superimposed with a complex BPSK signal having a carrier frequency $F_{\text{BPSK}} = 1.6 \text{ MHz}$. The BPSK signal is generated using random information bits at a rate of 5,000 bits per second, along with its Hilbert transform. In Figure 5a, which depicts the composite (chirp + BPSK + white Gaussian) signal for $\text{ISR} = 10 \text{ dB}$, and in Figure 5b, which depicts its time integration, the BPSK signal is shown to introduce pronounced interference near its carrier frequency with an integrated magnitude much larger than that of the chirp RFI owing to its time continuity.

Identical parameter settings for ISR , M/N , N , and L are used to construct ROC curves associated with the detection of the chirp + BPSK signal interferences; these results are depicted in Figure 6 following the format of Figure 4. Trends in chirp + BPSK detection performance are similar to those for chirp detection alone,

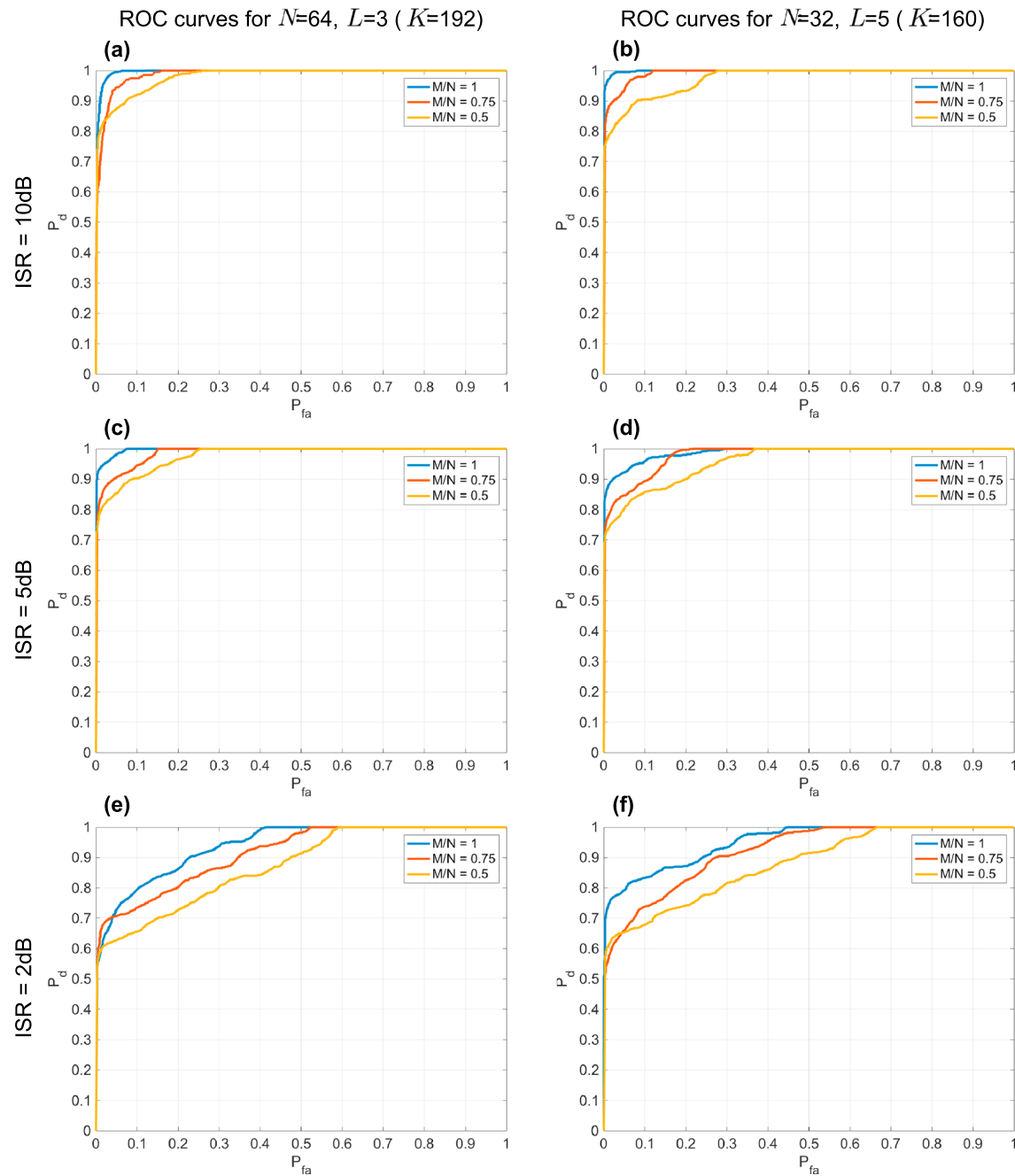


Figure 6. ROCs for the compressive statistical sensing-based cyclic feature detector for the case where the input signal is a complex chirp together with a BPSK signal centered at 1.6 MHz and additive white Gaussian noise. Format follows Figure 4. ISR = interference to signal ratio; ROC = receiver operating characteristic.

and, as before, high probabilities of detection are found for $p_{fa} < 0.2$ for all cases in which $ISR > 5$ dB. Detection of weak RFI, as depicted in Figures 6e and 6f, is rendered significantly worse by the added presence of BPSK interference (cf. against Figures 4e and 4f), such that high detection probabilities $p_d > 0.9$ are generally associated with relatively large false alarm rates $p_{fa} \approx 0.4$, an effect made worse with increasing compression ratio.

For the blanking procedure, we set $p_{fa} = 0.2$, which yields a threshold $\xi = 0.039$. Figure 5c depicts the resulting spectrogram of the signal after blanking, while Figure 5d shows its time integration. The shaded region indicates the BPSK frequencies over which the algorithm determined the signal to be fully contaminated.

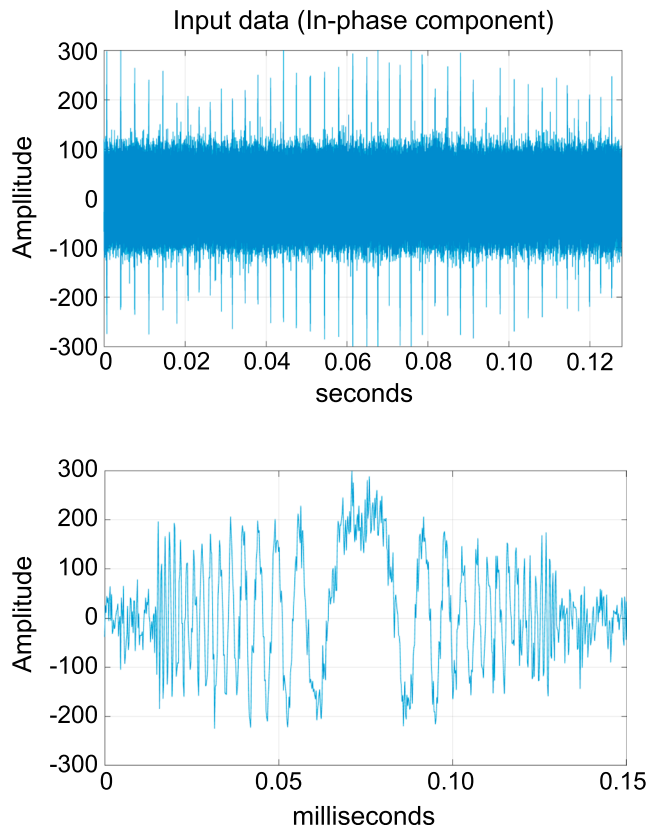


Figure 7. Real L-band signal acquired from Arecibo Observatory and used as input to the compressive statistical sensing-based algorithm. (top panel) In-phase component of the input signal for 0.12 s, it presents continuous radio frequency interference contamination generated by the active radar located near Arecibo Observatory facilities. (bottom panel) Plot of one pulse with duration 0.15 ms that shows the frequency-sweep pattern.

3.2. RFI Detection and Mitigation of Real Data

In this section, we examine actual raw data from the AO L-wide band antenna (dual polarization) centered at 1332.6 MHz with a 5-MHz bandwidth, which was acquired near noon on 17 May 2018. The data are contaminated by RFI mainly generated by the Punta Borinquen Common Air Route Surveillance Radar-CARSR (18°N, 67°W), whose four discrete frequency bands frequency hop across the L-band from 1269.41 to 1332.6 MHz. The measured signal is downconverted to baseband and centered to 0 Hz with a final frequency range $[-2.5, 2.5]$ MHz. Figure 7 depicts the in-phase component of the input signal used in this study: The top panel shows a 0.12-s-length signal that presents continuous pulse-like RFI contamination, while the bottom panel depicts one pulse (0.15-ms duration) extracted from the data shown in the top panel, where the frequency-sweep pattern is evident.

As with to the procedure to detect synthetic RFI described in section 3.1, our CSS-based algorithm was applied to the actual AO data using the following parameters: $K = 192$, $N = 64$, $L = 3$, and $M/N=0.75$, since they exhibited the best performance in our synthetic experiments. In order to validate this selection, we conducted an additional subexperiment using $K=128$, $N=64$, and $L=2$, where we evaluated the quality of the estimated cyclic spectrum. The result and conclusions can be seen in the supporting information Figure S2 and Text S2. This experimental study shows that our selection of number of samples, value of N , and L are sufficient to detect this specific chirp-like RFI signal in radar data. Figure 8 presents the estimation of the cyclic spectrum $\hat{S}_x$ from sequential pulses within the complex data $x[m]$. For each of the six graphs, identified with letters (a) to (f), the complex samples (in-phase and quadrature components) are shown on the left and the corresponding estimated 2-D cyclic spectrum $\hat{S}_x$ is shown on the right. The K -length window slides through the data, yielding Sequential Segments (a) to (f) such that Figure 8 depicts the behavior of the cyclic spectrum as the window passes by a pulse. For example, in Graph (a), we observe a peak value approximately in the pair

$(\alpha, f) = (0, -0.5)$ MHz, and their correspondent replicas in $(\alpha, f) = (1, 0)$ MHz and $(\alpha, f) = (-1, 0)$ MHz. The last two pairs indicate the cyclostationary condition of the found frequency. The following graphs, (b) to (f), show the displacement of this peak from negatives to positives frequencies. Cyclic spectra are limited to the ranges $f \in [-1.5, 1.5]$ MHz and $\alpha \in [-3, 3]$ MHz for better visualization and amplitude is set to 0.05 to better compare the results.

An example result of blanked RFI in the real AO data is depicted Figure 9, obtained using an experimental threshold of $\xi=0.026$. Figure 9a shows the spectral content of a representative portion of the analyzed data with a duration of ≈ 4 ms, which contains two pulses, while Figure 9b shows its time integration. Figures 9c and 9d depict the results of the blanking process, where the algorithm demonstrates its capability to determine the presence of RFI signatures. Frequency channels from 20 to 40, associated with a frequency range $[-1, 1]$ MHz, are cleared of RFI contamination; however, the algorithm fails to detect the RFI present on the boundaries, that is, at frequencies greater than 1 MHz and lower than -1 MHz. The transient nature of the RFI signal in this region, which is around 10 samples and can be seen in the spectrogram because of the higher overlapping factor, precludes effective detection.

4. Discussion

In this study, we have used data from the AO L-wide receiver, which is contaminated with chirp-like short-duration RFI generated from an airport surveillance active radar. Such a chirp-like pattern has been previously studied and classified as a generalized almost cyclostationary (GACS) signal. Signals belonging to this class exhibit multivariate functions that are almost-periodic functions of time, whose Fourier series

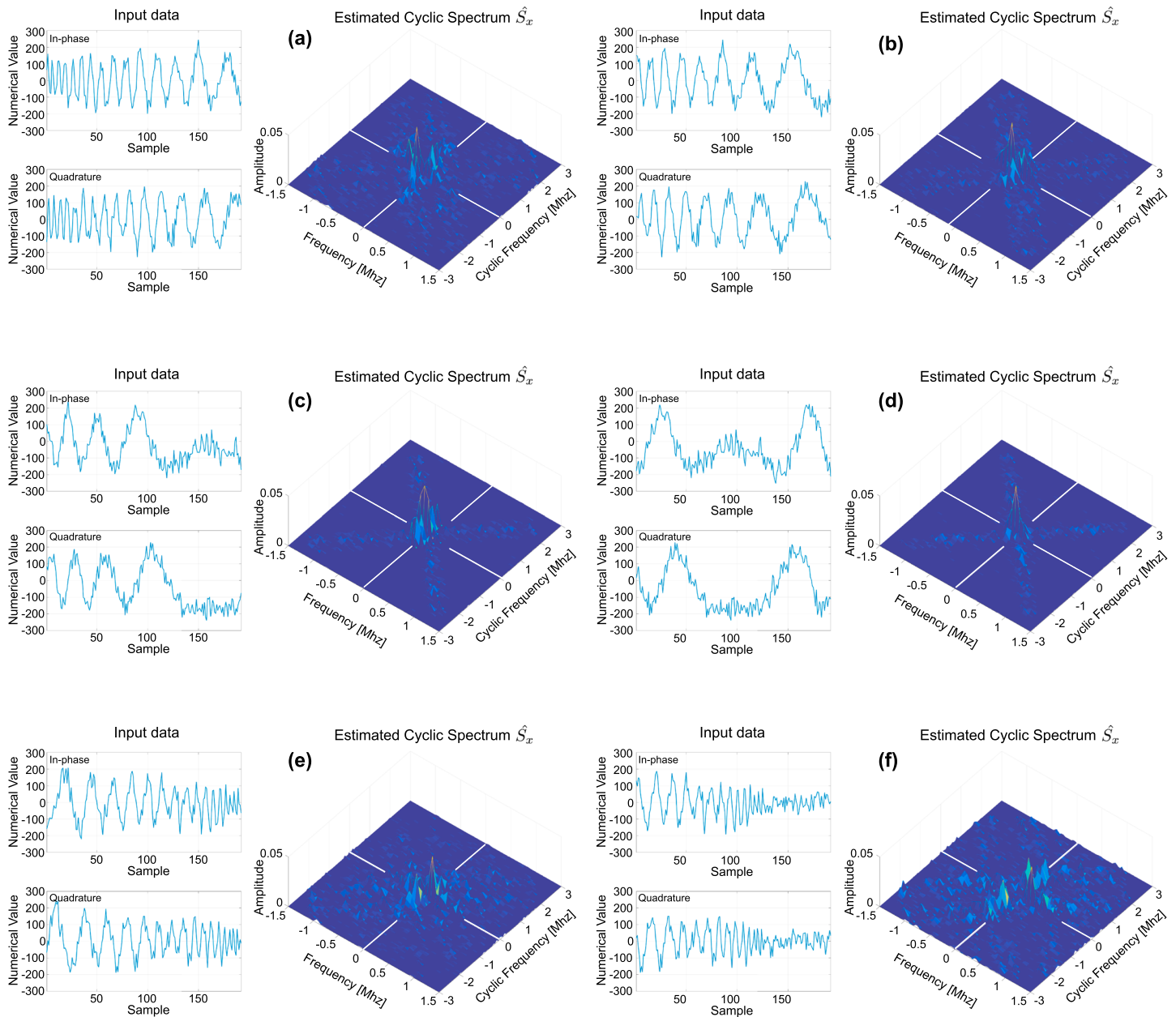


Figure 8. Estimated 2-D cyclic spectrum $\hat{S}_x$ over real data. Six graphs, (a) to (f), depict the estimation of the 2-D cyclic spectrum for a given complex input data $x[m]$, $x \in \mathbb{C}$. For each graph, $K = 192$ complex samples (in-phase and quadrature) used in the experiment are shown on the left and the estimated 2-D cyclic spectrum $\hat{S}_x$ is shown on the right. Input data sequence follows the alphabetic order (a) to (f). Cyclic spectra are limited to the ranges $f \in [-1.5, 1.5]$ MHz and $\alpha \in [-3, 3]$ MHz for better visualization. Amplitude is set to 0.05 to compare the results.

expansions have coefficients and frequencies that can depend on the lag shifts of the time series (Izzo & Napolitano, 1998, 2005). Several studies have investigated GACS signals and further assumed them as cyclostationary, such that techniques based on cyclic spectrum analysis can be used to estimate internal parameters of given GACS signals. For example, Gini et al. (2000) have estimated chirp radar signal parameters such as Doppler shift and Doppler rate by using a cyclostationary-based estimator. In addition, and for further details, the reader is referred to the section 17.3.2 in Madisetti (2010), where a cyclostationary test of a chirp signal is carried out. Our RFI detection algorithm is based on the general assumption of the cyclostationary of chirp signals, which has been validated by its good detection performance achieved using both synthetic and real data. Further steps, such as the identification of chirp

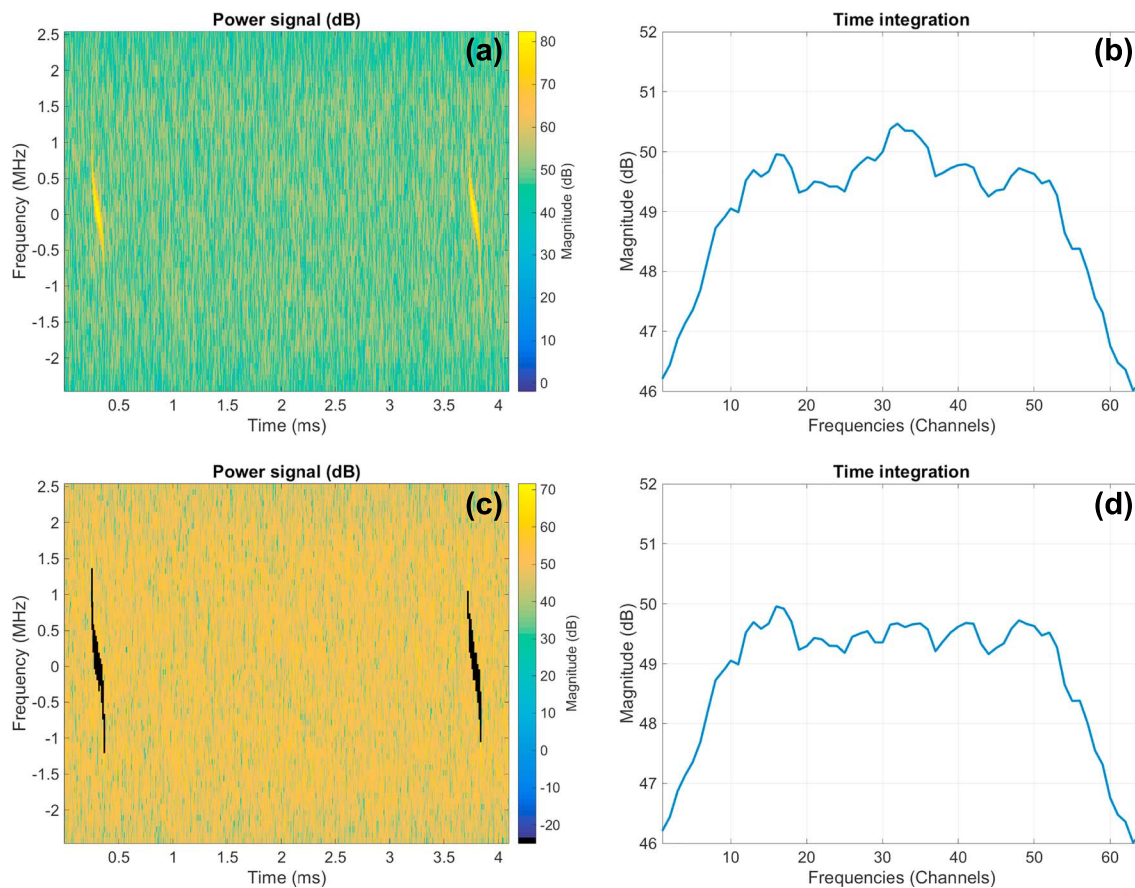


Figure 9. Spectrograms and time integration of a 4-ms segment of the real Arecibo Observatory L-band input signal, before and after detection and blanking of chirp-like radio frequency interference that is generated by nearby active radar. Format follows Figure 3.

parameters, could eventually be used for RFI modeling and potentially superior canceling (e.g., through direct subtraction).

In section 3.1, results with synthetic data have shown that our algorithm can determine the correct frequency content of a signal using only its sub-Nyquist samples. Moreover, our test of the algorithm's sensitivity to analysis parameters reveal that, even in an adverse scenario such as low strength RFI signal ($\text{ISR} = 2 \text{ dB}$), the algorithm can provide a probability of detection $p_d \sim 0.75$ for a compression ratio of $M/N = 0.5$. Although the radar signal in our analysis of actual AO L-band data possess a high ISR and the algorithm produces successful detection, our analysis of synthetic signals suggests that the algorithm could be used to detect other types of cyclostationary signals, such as BPSK, QPSK, among others, with low ISR that can be present in a single antenna system.

This research aims to contribute to real-time RFI detection and mitigation; however, further analysis should be done to determine the actual performance of the algorithm under a hardware-based implementation. Although such an investigation is beyond scope of the work reported here, we established the initial steps by introducing signal processing techniques, in the context of radio astronomy, that decrease the computational load. We make use of the CSS technique to estimate the cyclic spectrum of the radar RFI signal based on a reduced number of samples. As described in section 2, the CSS-based algorithm has two main procedures: generation of the intermediate matrix $\mathbf{T}$ through matrix-vector and matrix-matrix operations and the recovery of the cyclic spectrum $\hat{\mathbf{s}}_x$ through ℓ_1 -regularization. For the latter, we selected the FISTA algorithm since it performs iterative steps also based on simple matrix-matrix operations with a decreased convergence rate. In this context, current parallel architectures such as graphical processing units, which can handle thousands of cores, could leverage such structural operations to attain almost real-time results.

5. Conclusion

In this manuscript, we introduce a novel technique for RFI detection and mitigation based on CSS. Since man-made RFI have inherent periodicities involved in their formation, cyclostationarity analysis enables its detection and characterization in the frequency domain. CSS aims to estimate the cyclic spectrum of a signal by using sub-Nyquist samples and thus reducing the processing time. To assess our proposed algorithm, we have generated synthetic data with similar characteristics to man-made RFI generated by airport surveillance radar. ROC curves are used to evaluate the detection probability of the method. Successful detection is also observed in the analysis of real data from AO L-wide receiver contaminated with chirp-like RFI from airport surveillance active radar located near AO facilities. The promising ROC performance achieved with a data compression ratio of 0.5 along with short input time segments implies that short processing time allows for rapid estimation of frequency content as well as that the burden of high sampling rates required by high-dimensional statistical analysis of radio astronomy data can be relaxed to save acquisition costs.

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